CRISP-DM: Modeling

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1 Select Modeling Technique

The following models are selected to fulfill the data mining and business objectives:

1. **Logistic Regression**: the simplest and highly interpretable model for classification, though requires scaling the input data.
2. **Decision Tree**: the non-linear modeling tool with high explainability.
3. **Random Forest**: the ensemble approach to leverage the performance of each decision tree.
4. **Gradient Boosting**: the most powerful approach in the modeling of tabular data.

In addition, `sklearn` calibrated versions of the decision tree and random forest models are built.

1.1 Model Assumptions

- Logistic Regression: assumes linear relationship between log-odds and features; no multi-collinearity; independence of errors.
- Decision Tree / Random Forest / XGBoost: no distributional assumptions; can capture non-linearity and interactions; random forest assumes that trees are decorrelated.

2 Generate Test Design

Recall the data mining goals and their success criteria:

- ROC-AUC > 0.8 with at least 3 features having importance $\geq 1\%$ of the maximum importance.
- Calibration: Brier score < 0.1 and expected calibration error (ECE) < 0.1.
- Precision > 0.6 on the test set (at the threshold tuned on validation data), with recall expected in the range 0.25 ± 0.05.

The following procedures define how each test will be conducted. Actual results are reported in the *Assess Model* section.

ROC-AUC Test

- **Goal**: Measure ranking ability.
- **Method**: Compute ROC Area Under the Curve (ROC-AUC) on the test set.
- **Criterion**: ROC-AUC score > 0.8.
- **Data**: Test set.

Interpretability Test

- **Goal**: Identify features that contribute most to predictions and verify that at least three features have importance $\geq 1\%$ of the maximum importance.
- **Method**:
  - compute SHAP values on the test set and use mean absolute SHAP.
  - A feature is considered *important* if its importance $\geq 0.01 \times \max(\text{importance})$.
- **Criterion**: at least 3 features are identified.
- **Data**: Test set (mature developers).

Calibration Test

- **Goal**: Ensure predicted probabilities are reliable.
- **Method**:
  - Compute Brier score (mean squared error between probabilities and binary outcomes).
  - Compute Expected Calibration Error (ECE) with 10 equal-width bins.
  - Plot calibration curve (reliability diagram).
- **Criterion**: Brier score < 0.1 and ECE < 0.1.
- **Data**: Test set (mature developers).

Precision and Recall on Test Set

- **Goal**: Evaluate performance at the chosen decision threshold.
- **Method**:
  - Apply the selected threshold for each model to test set predictions, compute precision and recall.
- **Criterion**: precision > 0.6, recall within $[0.20, 0.30]$.
- **Data**: Test set (mature developers).

Recall on Top-Partnered Emerging Developers

- **Goal**: Verify transferability to the target population.
- **Method**:
  - From the emerging developers set (first release ≤ 3 years), keep those with `is_top` $= 1$.
  - Apply the same model and threshold used for the test set.
  - Compute the recall score.
- **Criterion**: A drop $< 50\%$ relative to test recall score.
- **Data**: Emerging developers set (not used for training).

3 Build Model

The following steps are performed for each model:

1. Data preparation.
2. K-Fold (K=5) hyperparameter tuning on the train set (80% of mature developers data from data preparation stage) by ROC-AUC score:
   - Cross-validation;
   - Selection of the best model;
   - Building the best model.
3. Selection of the best threshold that maximizes precision of a model: using out-of-fold set predictions (predictions made by the best model on K validation folds), with minimum recall of 0.25).

The sections below present the best parameters found by cross-validation ROC-AUC scores for each model (except for calibrated models). Moreover, Table 5 summarizes the best classification thresholds found by cross-validation precision scores for each model.

3.1 Logistic Regression

Table 1: Logistic Regression: Best Parameters and Validation ROC-AUC

Parameter	**Value**
C	0.1
penalty	l1
solver	liblinear
max_iter	1000
Best CV ROC-AUC	0.7805

3.2 Decision Tree

Table 2: Decision Tree: Best Parameters and Validation ROC-AUC

Parameter	**Value**
max_depth	5
min_samples_split	2
min_samples_leaf	1
max_features	None
criterion	entropy
class_weight	balanced
Best CV ROC-AUC	0.8160

3.3 Random Forest

Table 3: Random Forest: Best Parameters and Validation ROC-AUC

Parameter	**Value**
n_estimators	300
max_depth	None
min_samples_split	2
min_samples_leaf	4
max_features	sqrt
bootstrap	True
class_weight	balanced
n_jobs	-1
Best CV ROC-AUC	0.8698

3.4 Gradient Boosting (XGBoost)

Table 4: XGBoost: Best Parameters and Validation ROC-AUC

Parameter	**Value**
n_estimators	300
max_depth	7
learning_rate	0.01
subsample	0.8
colsample_bytree	0.8
scale_pos_weight	1
Best CV ROC-AUC	0.8703

3.5 Classification Thresholds

Table 5: Optimal Thresholds and Precision Scores (Validation Set)

Model	**Optimal Threshold**	**Precision**
Logistic Regression	0.2525	0.3831
Decision Tree	0.8586	0.3918
Random Forest	0.6263	0.5834
XGBoost	0.4141	0.5939
Decision Tree (calibrated)	0.3636	0.4535
Random Forest (calibrated)	0.4444	0.5759

4 Assess Model

The evaluation follows a funnel of tests:

1. Interpretability Test;
2. Calibration test (Brier score, ECE);
3. ROC-AUC;
4. Precision and recall on the test set, including recall on top-partnered emerging developers.

4.1 ROC-AUC

Table 6: ROC-AUC scores on the test set.

Model	**ROC-AUC**	**Yes** (> 0.8)
Logistic Regression	0.803	Yes
Decision Tree	0.822	Yes
Random Forest	0.882	Yes
XGBoost	0.881	Yes
Decision Tree (calibrated)	0.848	Yes
Random Forest (calibrated)	0.882	Yes

As shown in Table 6, all models except Logistic Regression achieve a ROC-AUC well above the required threshold. Business interpretation: most models succeed in separating the developers between two classes (top vs. non-top).

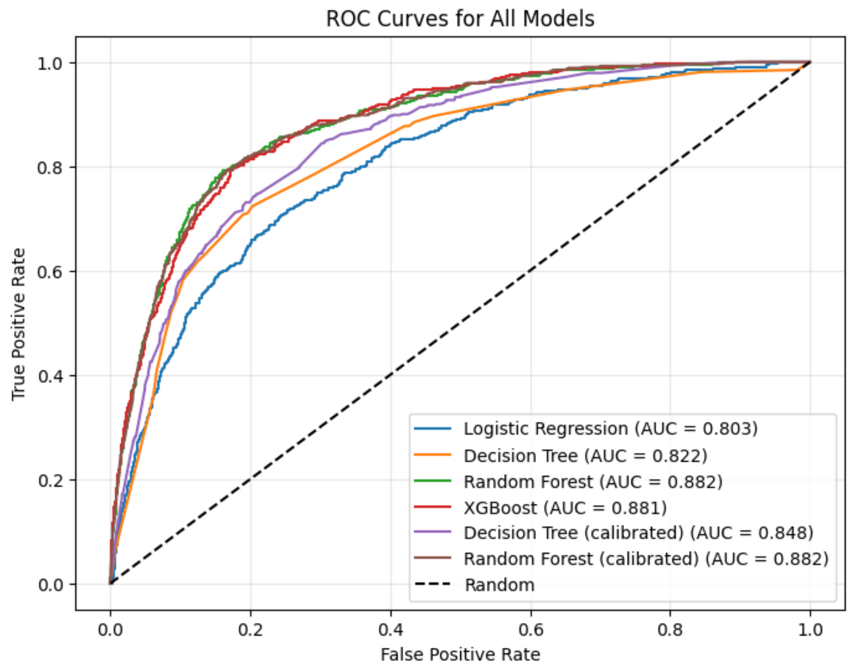

Figure 1: ROC evaluated on the test set.

4.2 Interpretability Test

Table 7: Number of significant features (importance $\geq 1\%$ of the maximum) per model.

Model	**Number of Important Features**	**Yes (≥ 3)**
Logistic Regression	17	Yes
Decision Tree	11	Yes
Random Forest	18	Yes
XGBoost	16	Yes
Decision Tree (calibrated)	11	Yes
Random Forest (calibrated)	18	Yes

As shown in Table 7, all models meet the criterion of having at least three features with importance above the 1% threshold. Business interpretation: all models provide interpretable results in terms of the developer portfolio.

4.3 Calibration Test

Table 8: Calibration metrics (Brier score and ECE) on the test set.

Model	**Brier Score**	**ECE**	**Yes (Brier<0.1, ECE<0.1)**
Logistic Regression	0.0777	0.0243	Yes
Decision Tree	0.1614	0.2536	No
Random Forest	0.0702	0.0678	Yes
XGBoost	0.0649	0.0194	Yes
Decision Tree (calibrated)	0.0708	0.0097	Yes
Random Forest (calibrated)	0.0640	0.0080	Yes

As shown in Table 8, all models except the uncalibrated Decision Tree are well calibrated. calibrated versions of Decision Tree and Random Forest show notable improvements. Business interpretation: most models provide reliable predictions that correspond to the real proportions of the top developers.

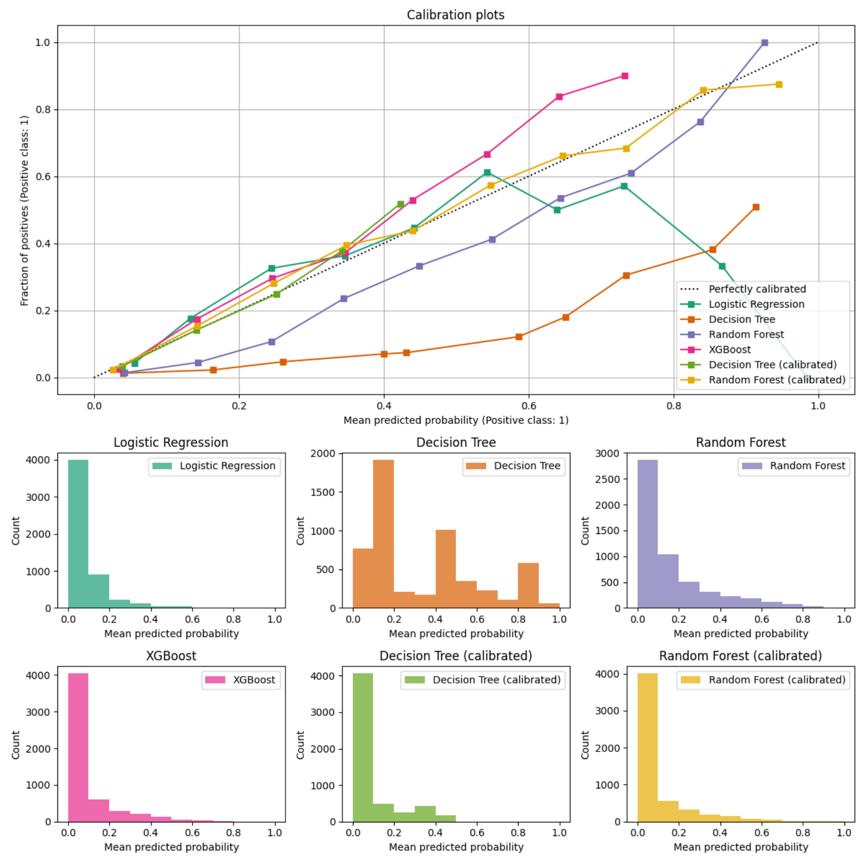

Figure 2: Calibration Results Plot

4.4 Precision and Recall on Test Set

Table 9: Precision and recall on the test set at the optimal threshold (tuned on validation data).

Model	**Precision**	**Recall**	**Yes (Prec**>0.6**, Recall** $\in [0.20, 0.30]$**)**
Logistic Regression	0.408	0.288	No
Decision Tree	0.385	0.358	No
Random Forest	0.623	0.248	Yes
XGBoost	0.631	0.246	Yes
Decision Tree (calibrated)	0.473	0.271	No
Random Forest (calibrated)	0.602	0.273	Yes

As shown in Table 9, Random Forest (uncalibrated and calibrated) and XGBoost meet the required precision and recall targets. Logistic Regression, Decision Tree (both versions) fail to achieve the required precision. Business interpretation: the three models are true in their predictions of top developers in at least 60% of cases and can detect 20% to 30% of top developers.

4.5 Recall on Top-Partnered Emerging Developers

Table 10: Recall on top-partnered emerging developers and drop relative to test recall.

Model	**Test Recall**	**Emerging Recall**	**Yes (Drop** $< 50\%$**)**
Logistic Regression	0.288	0.078	No (drop 73%)
Decision Tree	0.358	0.334	Yes (drop 7%)
Random Forest	0.248	0.128	Yes (drop 48%)
XGBoost	0.246	0.152	Yes (drop 38%)
Decision Tree (calibrated)	0.271	0.213	Yes (drop 21%)
Random Forest (calibrated)	0.273	0.144	Yes (drop 47%)

As shown in Table 10, all models except Logistic Regression maintain a recall drop of less than 50% when applied to top-partnered emerging developers. The Decision Tree exhibits the smallest drop, while Random Forest and XGBoost also demonstrate acceptable transferability. Business interpretation: most models have at least 50% of detection performance on top emerging developers, leading to higher chances to detect other promising emerging developers.